PH7002: Concepts in Statistical Mechanics
§2 Phase Transitions

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Aperitifs

Reading

Sethna sections 8.1, 8.3, 10.1, 10.2, chapter 11, and Yeomans chapter 1, sections 2,4, 2.5, 2.6, 3.1, 3.2.

In this set of notes, I will explain what are phase transitions and how do they come about. We will encounter many new concepts such as phase, phase boundary, thermodynamic limits, critical point, critical exponents, Maxwell construction, correlation functions. We will also encounter the famous Ising model, which is the archetypal model for studying phase transition.

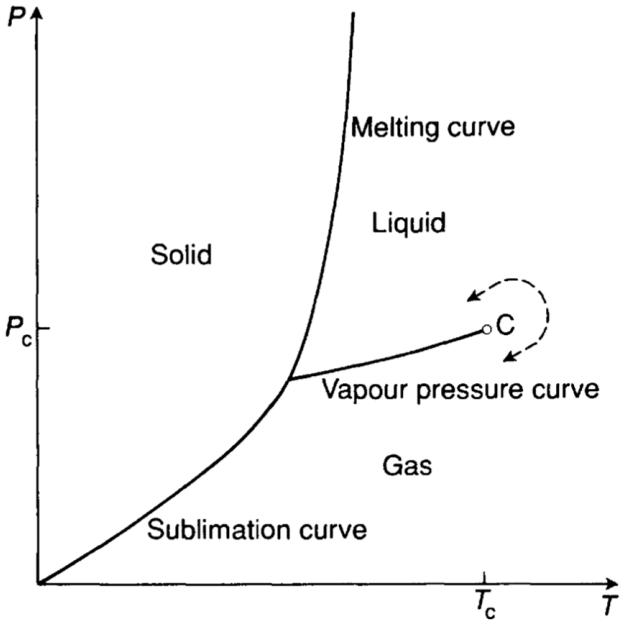
Important themes

1. New concepts such as phase, phase boundary, critical point, thermodynamic limit, critical exponents, correlation functions.
2. Van der Waals model and Maxwell construction.
3. Lee-Yang theory for phase transition.
4. Ising model.

Examples of phase transitions

The most spectacular consequence of interactions among particles is the appearance of new **phases** of matter whose collective behavior bears little resemblance to that of a few particles. How do the particles then transform from one macroscopic state to a completely different one? From a formal perspective, all macroscopic properties can be deduced from the free energy or the partition function. Since phase transitions typically involve dramatic changes in various response functions, they must correspond to **singularities** in the free energy. The canonical partition function for a finite collection of particles is always an analytical function. Hence phase transitions, and their associated non-analyticities, are only obtained for infinitely many particles, i.e. in the **thermodynamic limit**, $N \to \infty$. The study of phase transitions is thus related to finding the origin of various singularities in the free energy and characterizing them.

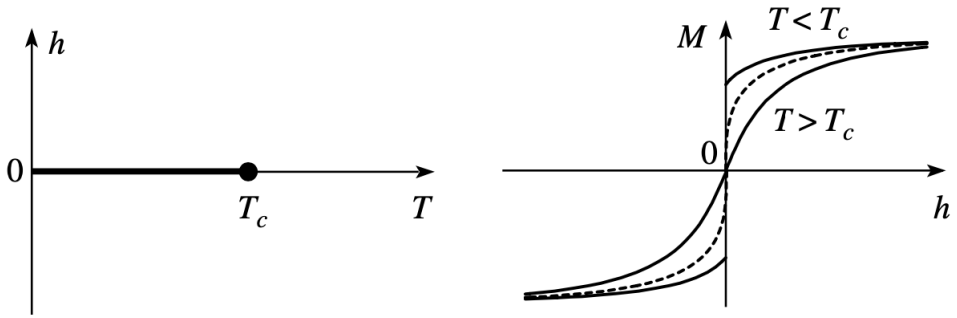
The classical example of a phase transition is the condensation of a gas into a liquid.



- In the temperature/pressure plane, (T, P), the phase transition occurs along a line that terminates at a critical point (T_c, P_c).
- Due to the termination of the **coexistence line**, it is possible to go from the gas phase to the liquid phase continuously (without a phase transition) by going around the critical point. Thus there are no fundamental differences between liquid and gas phases.

From a mathematical perspective, the free energy of the system is an analytical function in the (T, P) plane, except for a branch cut along the phase boundary.

A related, but possibly less familiar, phase transition occurs between paramagnetic and ferromagnetic phases of certain substances such as iron or nickel. These materials become spontaneously magnetized below a **Curie temperature** T_c. There is a discontinuity in magnetization of the substance as the magnetic field h goes through zero for $T < T_c$. The phase diagram in the (h, T) plane, and the magnetization isotherms $M(h)$, have much in common with their counterparts in the condensation problem.



In both cases a line of discontinuous transitions terminates at a critical point, and the isotherms exhibit singular behavior in the vicinity of this point. The phase diagram of the magnet is simpler in appearance, because the Z_2 symmetry $h \leftrightarrow -h$ ensures that the critical point occurs at $h_c = M_c = 0$.

Thermodynamic limit

Experience tells us that the free energy is extensive for a large system:

where f_{b} is the bulk free energy per unit volume or bulk free energy density. The precise definition is

For a system defined on a lattice, with N lattice sites, the bulk free energy per site is

The limit is known as the thermodynamic limit.

Sometimes, an auxiliary constraint is imposed simultaneously with the limit: for example, in fluid systems, taking the limit $V \to \infty$ is senseless unless one simultaneously takes the limit that the number of particles N in the system also tends to infinity, in such a way that the density N/V stays constant. We will see that the concepts of phase and and phase transition are only sharply defined in this limit. The existence of the thermodynamic limit is not trivial.

Phase boundaries and phase transitions

When f_{b} exists, then a precise definition of a **phase boundary** follows. Let us suppose that there are D coupling constants. The axes of the phase diagram are K_1, K_2, $\ldots K_D$, and hence the dimension of the phase diagram is D. As a function of $\{K_n\}$, $f_{\mathrm{b}}[K]$ is analytic almost everywhere. The possible non-analyticities of $f_{\mathrm{b}}[K]$ are points, lines, planes, hyperplanes, etc. in the phase diagram. These singular loci have a dimensionality associated with them ($D_s = 0, 1, 2, \ldots$ respectively), and the important invariant for each type of singular locus is the **codimension**, C:

This is an invariant in the sense that if we decide to include an extra variable in $\{K_n\}$ (and thus to the phase diagram), both D and D_s, increase by 1 so that C remains fixed.

Regions of analyticity of $f_{\mathrm{b}}[K]$ are called **phases**. Loci of codimension $C = 1$ (i.e. loci which separate phases), are called phase boundaries. Loci of codimension greater than one cannot possibly represent phase boundaries. To give a prosaic example, in order to partition a room into two, it is necessary to build a wall across it - a pole through the centre of the room will not suffice.

The definition of phase given above is ambiguous. There may exist a path along which $f_{\mathrm{b}}[K]$ is analytic going from one side of a phase boundary to the other. Using our "room analogy", this is like having a wall which does not quite reach the ceiling. At floor level, the room is partitioned, but a flying insect may be able to pass from one side of the room to the other without encountering any impediment to its progress.

An example is the liquid-gas-solid phase diagram. Although it is not possible to pass from fluid to solid without encountering a phase transition, it is possible to choose a path in $p - T$ space which goes from liquid to gas without encountering any singular behaviour in thermodynamic quantities.

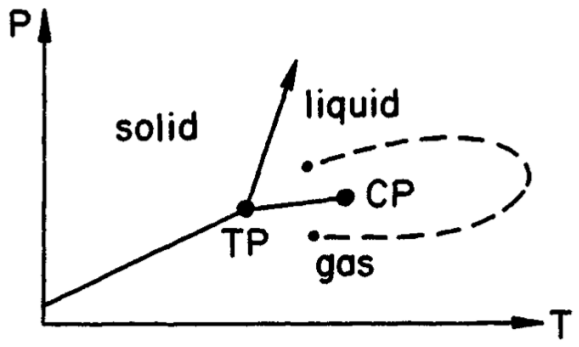


Figure: The point TP is a triple point ($C = 2$) and the point CP is a critical point ($C = 2$). The solid-liquid phase boundary extends to arbitrarily high pressure (in principle), whilst the gas-liquid phase boundary ends at CP. The dashed curve represents a trajectory in the phase diagram along which no phase transition is encountered, even though a change of phase has apparently taken place.

$f_{\mathrm{b}}[K]$ is everywhere continuous. This implies that where phase boundaries exist, they must come in two classes:

1. $\partial f_{\mathrm{b}}[K]/\partial K_i$ is discontinuous across a phase boundary. It can be either one or more of the $\partial f_{\mathrm{b}}[K]/\partial K_i$ which is discontinuous. If this case occurs, then the transition is said to be an **abrupt phase transition** (or first-order phase transition).
2. The only other remaining possibility for non-analytic behaviour is that all $\partial f_{\mathrm{b}}[K]/\partial K_i$ are continuous across the phase boundary. If this occurs, the transition is said to be a **continuous phase transition**, or second order transition (this terminology is not encouraged).

Critical behavior

The **critical exponent** associated with a function $F(t)$ is

It is more usually written as

Zero-field specific heat	$C_H \sim \mid t \mid^{-\alpha}$
Zero-field magnetization	$M \sim (-t)^{\beta}$
Zero-field isothermal susceptibility	$\chi_T \sim \mid t \mid^{-\gamma}$
Critical isotherm ($t = 0$)	$H \sim \mid M \mid^{\delta} \operatorname{sgn}(M)$
Correlation length	$\xi \sim \mid t \mid^{-\nu}$
Pair correlation function at T_c	$G(\vec{r}) \sim 1/r^{d-2+\eta}$

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Example 2.1

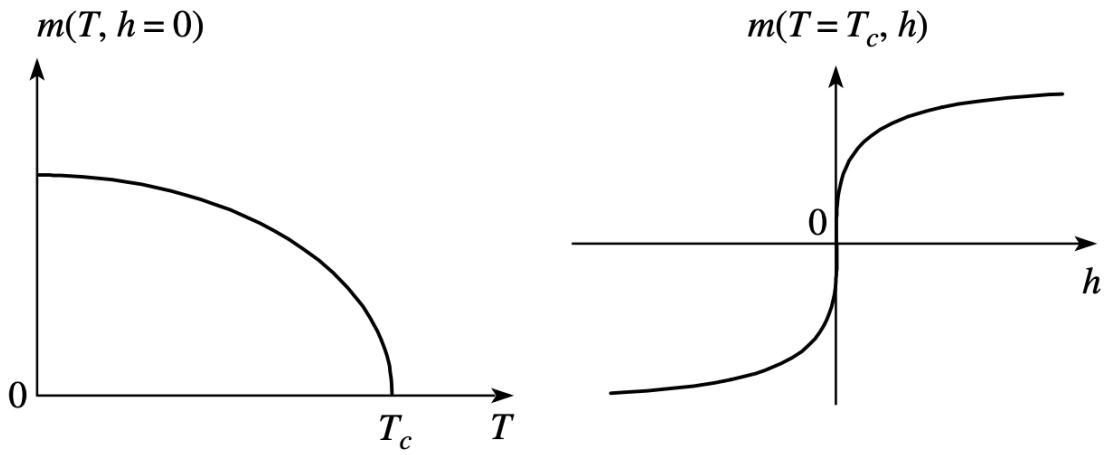
Yeomans problem 2.3 (i) and (iv).

2.3 Determine the critical exponents λ for the following functions as $t \to 0$:

$$
\begin{aligned}
&\text{(i)} && f(t) = At^{1/2} + Bt^{1/4} + Ct \\
&\text{(ii)} && f(t) = At^{-2/3}(t+B)^{2/3} \\
&\text{(iii)} && f(t) = At^2 e^{-t} \\
&\text{(iv)} && f(t) = At^2 e^{1/t} \\
&\text{(v)} && f(t) = A\ln\{\exp(1/t^4) - 1\}
\end{aligned}
$$

[Answers: (i)1/4, (ii)−2/3, (iii)2, (iv)undefined, (v)−4.]

The singular behavior in the vicinity of a critical point is characterized by a set of **critical exponents**. These exponents describe the non-analyticity of various thermodynamic functions. The most commonly encountered exponents are listed below:



[Video: Benoit Mandelbrot on fractals]

Order Parameter

By definition, there is more than one equilibrium phase on a coexistence line. The **order parameter** is a thermodynamic function that is different in each phase, and hence can be used to distinguish between them. For a magnet, the magnetization

In zero field, m vanishes for a paramagnet and is non-zero in a ferromagnet, and close to the transition behaves as

where $t = (T_c - T)/T_c$ is the reduced temperature.

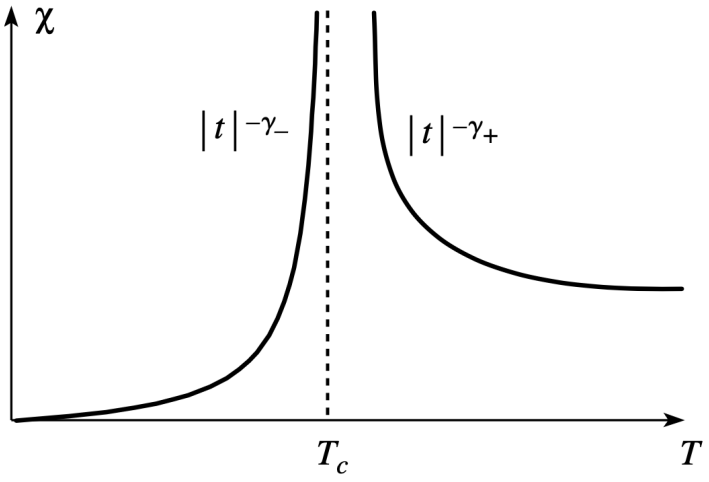
Non-analyticity

The singular behavior of the order parameter along the coexistence line is traditionally indicated by the **critical exponent** β. The phase transition disappears in a finite magnetic field, i.e. the magnetization varies continuously with T at a finite h. This **non-analyticity** is thus reminiscent of a branch cut in a complex plane. The magnetization is non-analytic for any trajectory in the (T, h) plane that passes through the critical point at $T = T_c$ and $h = 0$. In particular, it vanishes along the critical isotherm, with another exponent denoted by δ, defined through

Response functions

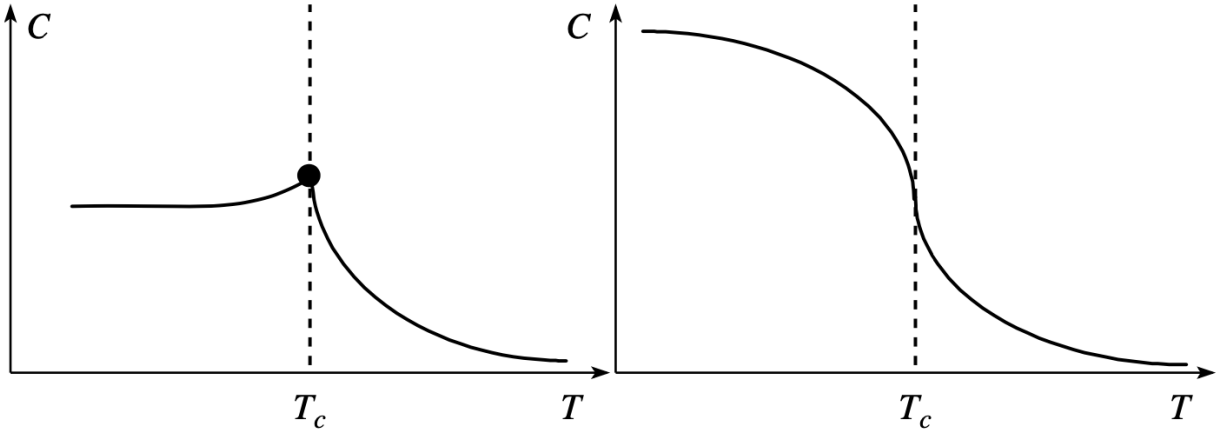
The critical system is quite sensitive to external perturbations, as typified by the **infinite** compressibility at the liquid–gas critical point. The **divergence** in the response of the order parameter to a field conjugate to it is indicated by an exponent γ. For example, in a magnet,

where in principle two exponents γ_+ and γ_- are necessary to describe the divergences on the two sides of the phase transition. Actually in almost all cases, the same singularity governs both sides and $\gamma_+ = \gamma_- = \gamma$.

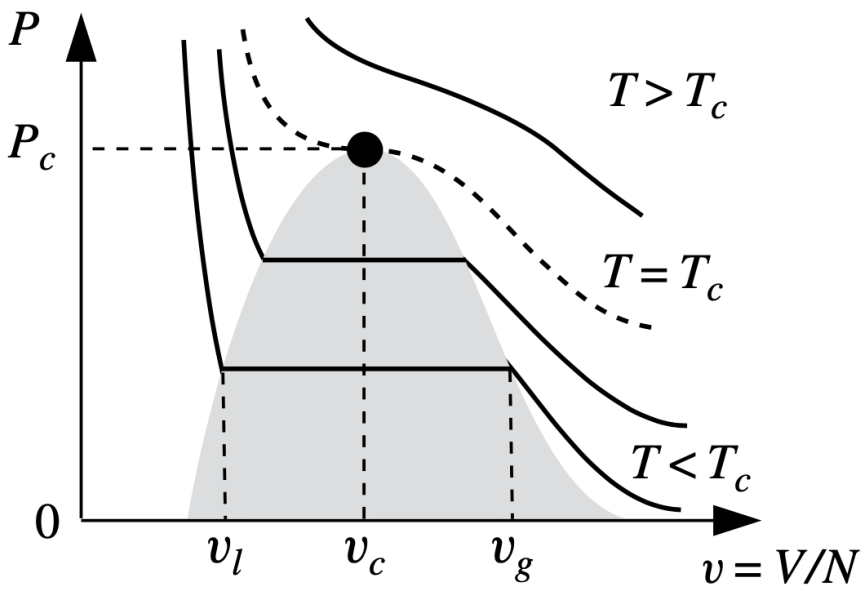


The heat capacity is the thermal response function, and its singularities at zero field are described by the exponent α, i.e.

A positive α corresponds to a divergence, while a negative α describes a finite heat capacity, possibly with a cusp. A negative heat capacity exponent α may (left) or may not (right) describe a cusp singularity depending on the relative signs of amplitudes.



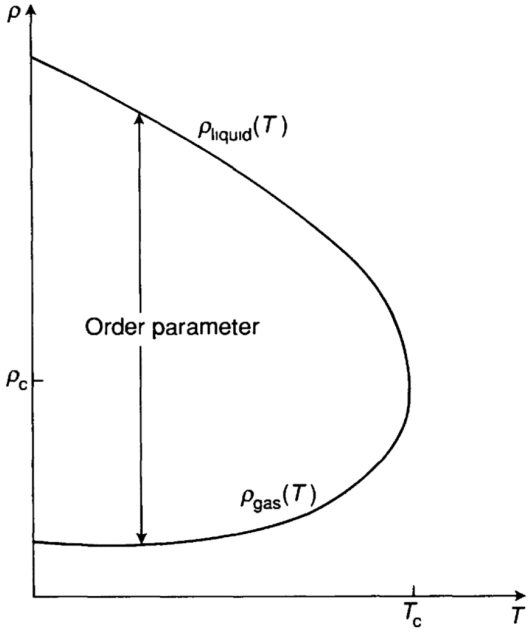
The two phases along the liquid–gas coexistence line can be differentiated by their densities ρ_g and ρ_ℓ. Their differences from the critical point density, $\rho_\ell - \rho_c$ and $\rho_c - \rho_g$ can serve as the order parameter. In the vicinity of the critical point, both these quantities, as well as $\rho_\ell - \rho_g$ vanish with the exponent β. The exponent δ now describes the singular form of the critical isotherm



- The pressure versus volume isotherms become progressively more flat on approaching T_c from the high temperature side. This implies that the isothermal compressibility,

  diverges as $T \to T_c^+$.
- The fluid appears "milky" close to criticality. This phenomenon, known as critical opalescence, suggests that collective fluctuations occur in the gas at long enough wavelengths to scatter visible light. These fluctuations must necessarily involve many particles, and a coarse-graining procedure may thus be appropriate to their description. Videos: Experiment demonstrating Critical Opalescence, Benoit Mandelbrot on Critical Opalescence.

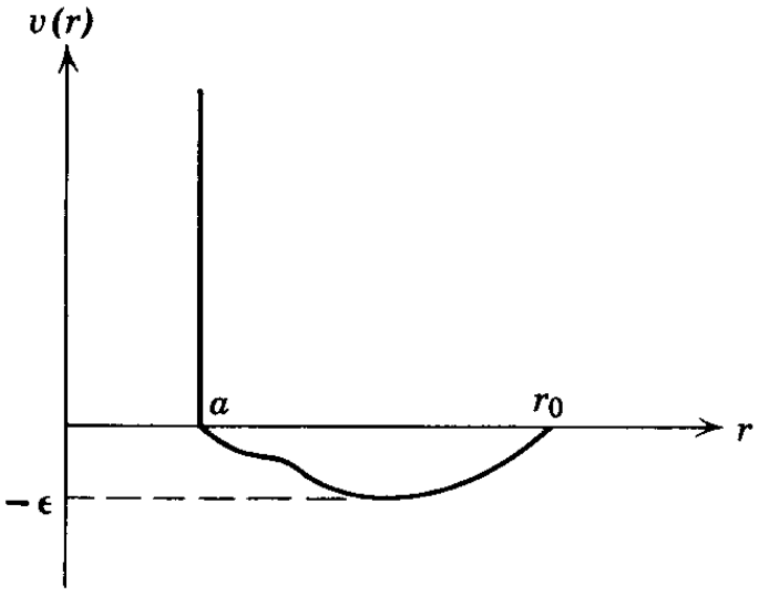
The difference between the densities of coexisting liquid and gas phases vanishes on approaching T_c, i.e. $\rho_\ell \to \rho_g$, as $T \to T_c^-$.



Lee-Yang Theory

All properties of the interacting system, including phase separation, are contained within the thermodynamic potentials that can be obtained by evaluating the partition function. Phase transitions, however, are characterized by discontinuities in various state functions and must correspond to the appearance of singularities in the partition functions. At first glance, it is somewhat surprising that any singular behavior should emerge from computing such well-behaved integrals.

T.D. Lee and C.N. Yang (Nobel Prize in Physics 1957) proposed a definite scenario for the occurrence of singularities in the thermodynamic limit. Consider N molecules in volume V, interacting with one another through a pairwise potential.



Each molecule is taken to be a hard sphere surrounded by an attractive potential of finite range. Thus, a finite volume V can accommodate at most $M(V)$ molecules. For $N > M(V)$, the partition function vanishes because at least 2 molecules will touch, rendering the energy infinite.

The grand partition function is a polynomial of degree $M(V)$ in the fugacity z

Since all coefficients $Z_N(V)$ are positive, the polynomial has no real positive roots.

The parametric form of the equation of state is

For any finite V, both P and v are analytic function of z in a region of the complex z plane that includes the entire real axis. Therefore P is an analytic function of v in a region of complex v plane that includes all physical values of v, i.e. the real axis. Hence, all thermodynamic functions must be free of singularities. We require $P > 0$ and

where n is the density.

To have singularities, we must go to the thermodynamic limit $V \to \infty$ at fixed v

Note that, in the 2nd equation, the order of the operations, lim and $z(\partial/\partial z)$ can be interchanged only if the limits above are approached uniformly. Yang and Lee show that phase transitions are controlled by the **distribution of roots** of the grand partition function in the complex z plane. A phase transition occurs whenever a root approaches the **real axis** in the limit $V \to 0$.

Theorem 1

The limit

exists for all $z > 0$, and is continuous nondecreasing function of z. This limit is independent of the shape of V, if the surface area of V increases no faster than $V^{2/3}$.

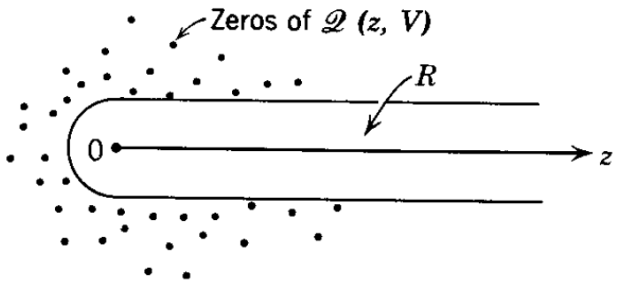
Theorem 2

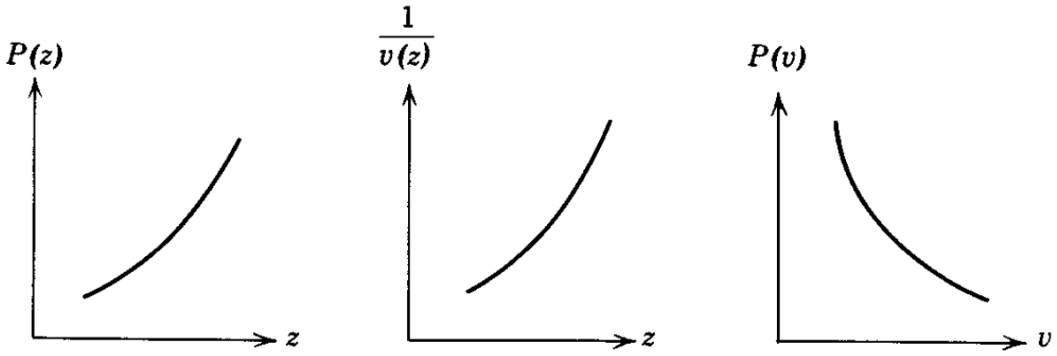
Suppose R is a region in the complex z plane that includes a segment of the positive real axis, and contains no roots of the grand partition function. Then in this region, the quantity $V^{-1} \log \Xi$ converges uniformly to its limit $V \to \infty$. The limit is analytic for all z in R.

A thermodynamic phase is defined by those values of z contained in any single region R of theorem 2. Since in any region R the convergence to the limit $F_\infty(z)$ is uniform, we can interchange the order of lim and $z(\partial/\partial z)$. In any single phase, the equation of state is given in parametric form by

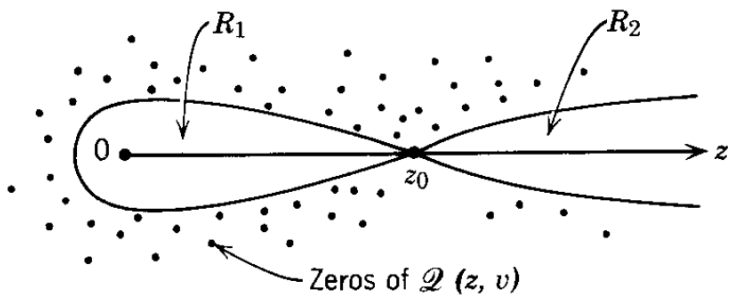
The properties $P > 0$ and $\partial P/\partial v < 0$ are maintained in the thermodynamic limit.

Suppose the region R includes the entire real positive z axis and is free of zeroes of $\Xi(z, V)$. Then the system always exists in a single phase. The equation of state may be obtained by eliminating z.



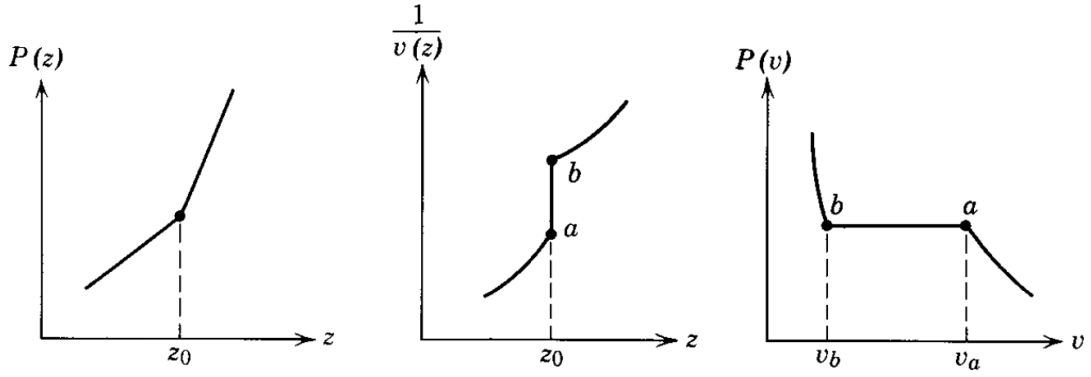


A zero of the grand partition function approaches z_0 on the real positive z axis that seperates two regions R_1 and R_2, each free of zeroes of $\Xi(z, V)$. At z_0, $P(z)$ must be continuous, as required by theorem 1. However, its derivative may be discontinuous. The system then possesses two phases, corresponding to the region $z < z_0$ and $z > z_0$. The two phases are connected by a phase transition.

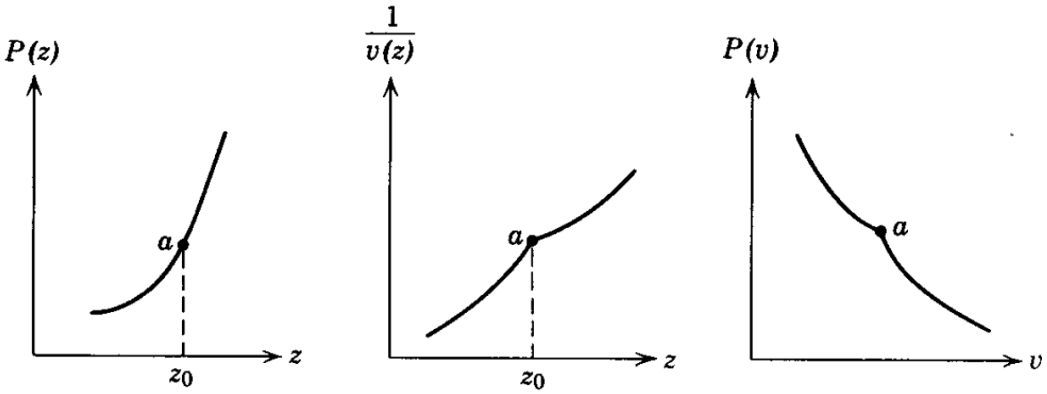


Now, $1/v(z)$ is a nondecreasing function of z

If $\partial P/\partial z$ is discontinuous, $1/v(z)$ must make a discontinuous upward jump when z increases through z_0. This results in a first-order phase transition or abrupt phase transition. The fact that $1/v(z)$ actually assumes all values between the points a and b follows from the fact that the curve of $1/v(z)$ is the limiting curve of a sequence of continuous curves.



If $\partial P/\partial z$ is continuous at $z = z_0$, but $\partial^2 P/\partial z^2$ is discontinuous, then we have a second-order phase transition.



Lee-Yang Circle Theorem

Example 2.2

Lattice gas with attractive interactions considered by Lee and Yang.

Consider a lattice gas with attractive interactions. The system consists of point atoms located on the sites of a discrete lattice, with the condition that no two atoms occupy the same site, and that atoms on different sites have attractive pairwise interactions. The interatomic potential u is such that

The nature of the interactions and dimensionality or structure of the lattice can be arbitrary.

The **Lee-Yang circle theorem** states that, for the lattice gas model, all roots of the equation

lie on the unit circle in the complex z plane. For finite V, the roots occur in complex-conjugate pairs, and none can touch the real axis. Only in the limit $V \to \infty$ is it possible for any of them to approach the real axis at $z = 1$. Thus, there is at most one transition point.

Denoting the phase angles of the roots by θ_k, we have

In the thermodynamic limit, the roots become continuously distributed on the unit circle. We define a distribution function $g(\theta)$ as

The fact that roots occur in complex-conjugate pairs means that

Using this, we can deduce the equation of state

We can see from this general form that as z varies along the real axis, the only point where the singularities can occur is $z = 1$. A singularity at this point will occur if $g(0) = 0$. If $g(0) \neq 0$, then $z = 1$ will be a singular point and the equation of state will give different functions for $z > 1$ and $z < 1$, which cannot be analytically continued into each other, i.e. existence of 2 phases and transition at $z = 1$.

Ising model

Phase transitions provide the most striking manifestations of interparticle interactions. Our discussion of the microscopic theory for this type of phenomena will start with a simple lattice model that describes magnetism, known as the **Ising model**. We consider a system of N spins arranged on a lattice in the presence of a uniform magnetic field, H.

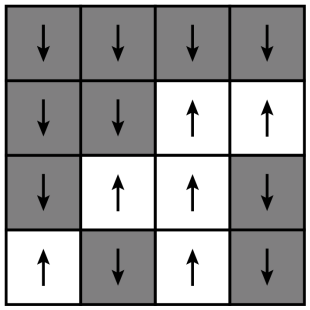
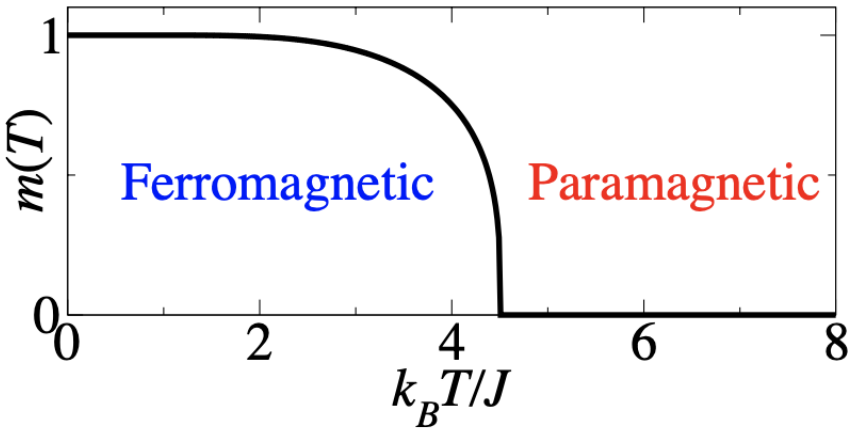


Fig. 8.1 The 2D square-lattice Ising model. It is traditional to denote the values $s_i = \pm 1$ as up and down, or as two different colors.

Notice that when $J > 0$, it is energetically favorable for neighboring spins to be aligned. Hence, we might anticipate that for low enough temperature, this stabilization will lead to a cooperative phenomenon called spontaneous magnetization. That is, through interactions between nearest-neighbors, a given magnetic moment can influence the alignment of spins that are separated from the given spin by a macroscopic distance. These long ranged correlations between spins are associated with a long ranged order in which the lattice has a net magnetization even in the absence of a magnetic field. The system at zero field displays Z_2 symmetry:

The magnetization

in the absence of the external magnetic field, H, is called the spontaneous magnetization.

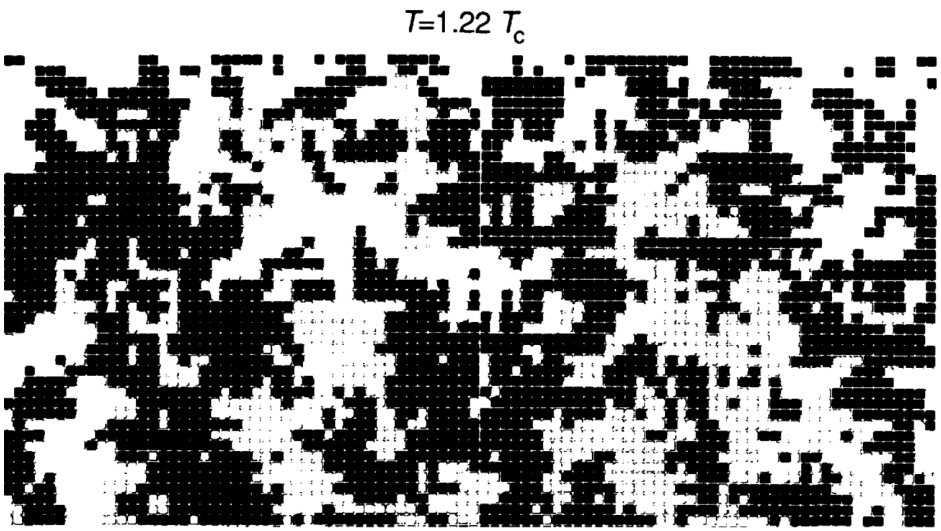


Unless the temperature is low enough (or J is high enough), there will be no net magnetization. Let T_c (the Curie temperature or critical temperature) denote the highest temperature for which there can be non-zero magnetization.

Example 2.3: Ising model

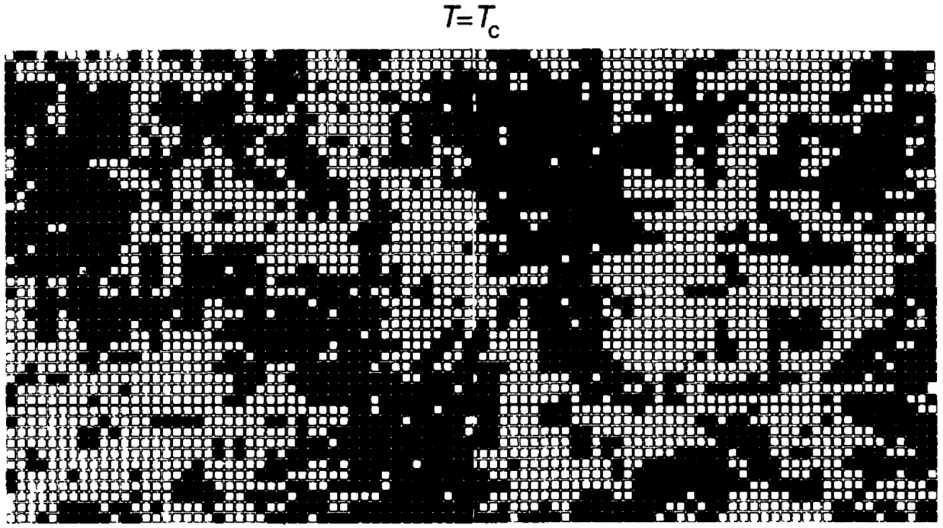
Sethna problem 8.1

Play with the online simulator: website

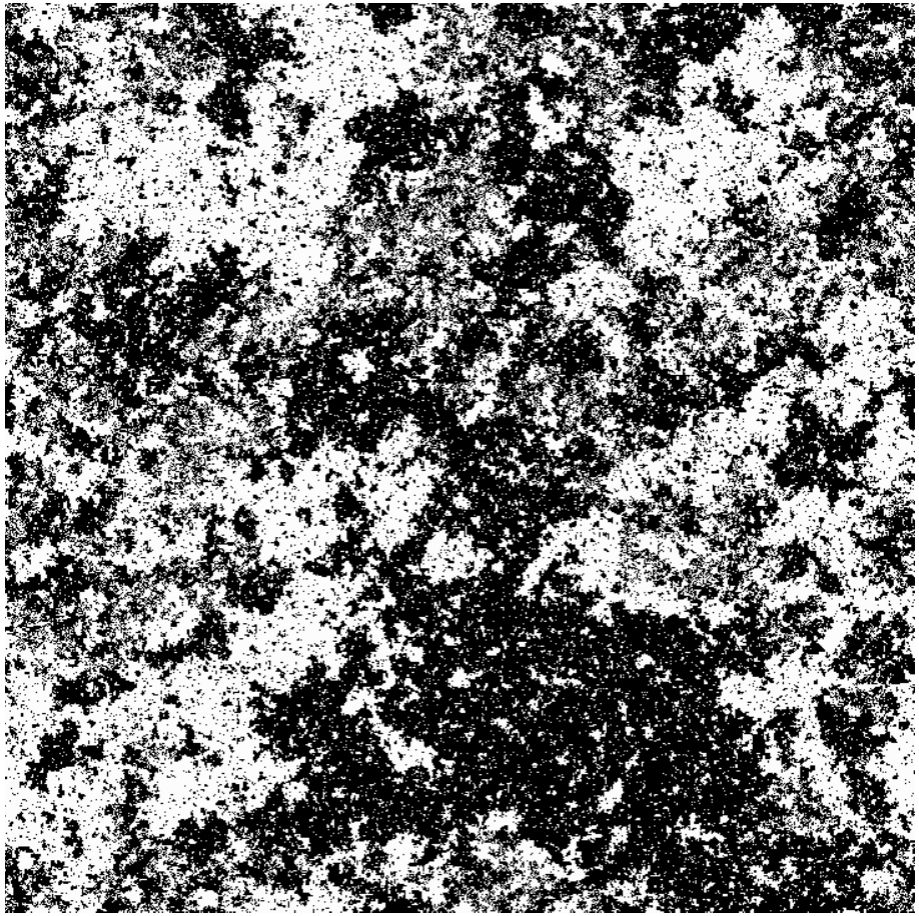


Note that $T_c = 2.269185$.

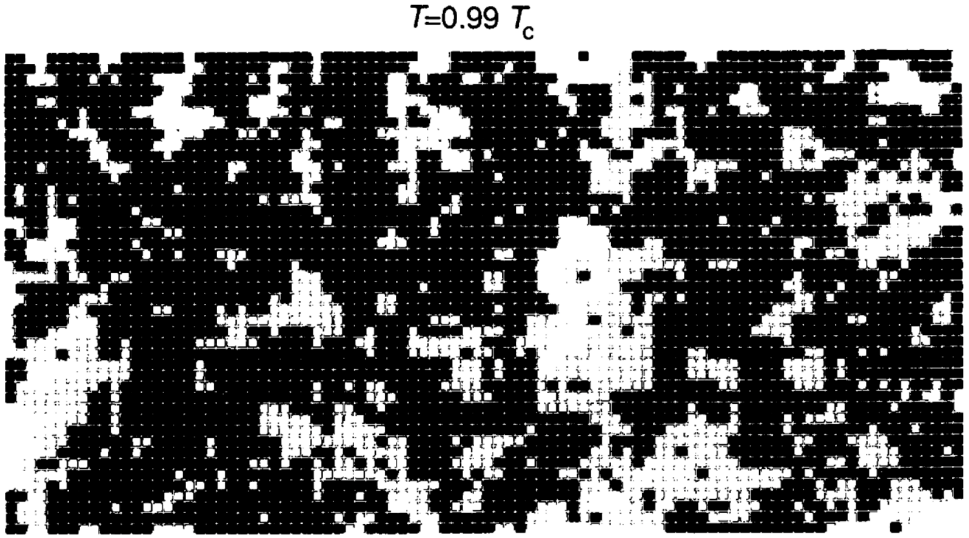
To understand what is happening microscopic, we can simulate the 2D Ising model on a computer. At temperature above T_c, entropic contributions dominate the exchange energy, and although nearest neighbors tend to lie parallel, there is a small perturbation on a random configuration. The size of the largest clusters is measured by its **correlation length**. At $T = T_c$, the correlation length is of the order of a few lattice spacings. The system is said to show **short-range order**.



As the temperature is lowered, the correlation length increases. Critically, fluctuations on a smaller scale remain important; there are correlated regions of spins on *all* length scales up to that set by the correlation length. Each fluctuation is not an area of uniform spin alignment but includes smaller fluctuations which in turn include yet smaller ones down to the length scale set by the lattice spacing. The critical temperature $T = T_c$ is marked by the correlation length becoming infinite. There is now no upper length cut-off and ordered structures exist on every length scale. **At a critical phase transition, fluctuations on all scales of length are important.**



The two-dimensional square-lattice Ising model near the critical temperature T_c. Here the 'islands' come in all sizes, and the equilibrium fluctuations happen on all time scales;



Below the critical temperature $T < T_c$, there is a non-zero magnetization. More spins lie in one of the two spin states. The model is said to exhibit **long-range order**. At zero temperature, all the spins are aligned because of the exchange interaction. As the temperature increases entropic terms in the free energy lead to fluctuations away from this state and the magnetization drops from its saturated value.

The correlation length measures the size of the largest fluctuations away from the ordered background. As the temperature increases towards the critical temperature, the correlation length becomes larger. Just as for $T > T_c$, there are clusters embedded within clusters on all length scales. The fluctuations cause the magnetization to fall, and it drops to zero exactly at the critical temperature where the correlation length becomes infinite and the underlying order is completely destroyed.

Correlation functions

A material in thermal equilibrium may be macroscopically homogeneous and static, but it wiggles on the microscale from thermal fluctuations. We measure how systems wiggle and evolve in space and time using correlation functions.

Figure: Ising model at low temperature, showing the spin $S(\mathbf{x}, t)$ at position $\mathbf{x}$ and time t; the up-spin and down-spin regions are competing to determine which will take over as the broken-symmetry ground state.

We typically measure the space and time coherence in a system using correlation functions. Consider the alignment of two Ising spins $S(\mathbf{x}, t)$ and $S(\mathbf{x} + \mathbf{r}, t)$ in the coarsening figure; spins at the same time t, but separated by a distance $\mathbf{r}$. If $|\mathbf{r}|$ is much larger than a typical blob size $L(t)$, , the spins will have a 50/50 chance of being aligned or misaligned, so their average product will be near zero. If $|\mathbf{r}|$ is much smaller than a typical blob size L, the spins will typically be aligned parallel to one another (both $+1$ or both -1), so their average product will be near one. The equal-time spin-spin correlation function

will go from one at $\mathbf{r} = \mathbf{0}$ to zero at $|\mathbf{r}| \gg L(t)$, and will cross $1/2$ at a characteristic blob size $L(t)$.

The correlation function in general contains more information than just the typical blob size. Consider the equilibrium correlation function, say for the Ising model

The equal-time correlation function $C(\mathbf{r}, 0)$ contains information about how much a spin influences its distant neighbors. Even at high temperatures, if a spin is up its immediate neighbors are more likely to point up than down. As the temperature approaches the ferromagnetic transition temperature T_c, this preference extends to further neighbors. Below T_c we have long-range order; even very distant neighbors will tend to align with our spin, since the two broken-symmetry equilibrium states each have net magnetization per spin m. Above T_c the equal-time correlation function goes to zero at long distances r; below T_c it goes to m^2, since the fluctuations of two distant spins about the mean magnetization become uncorrelated:

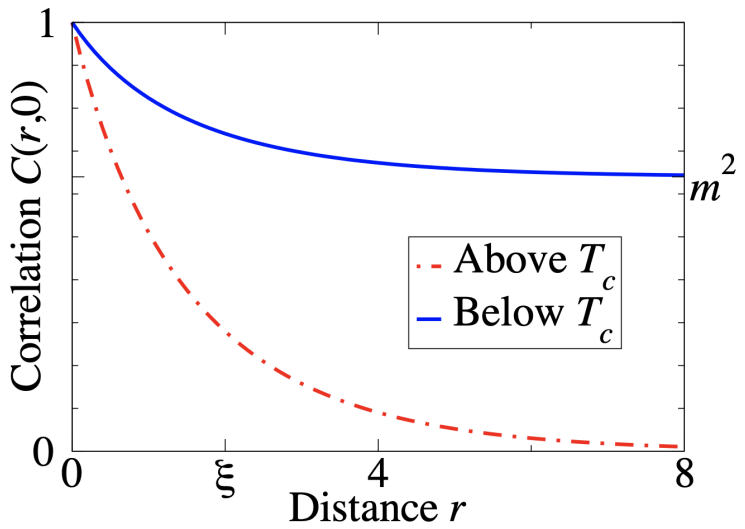


Figure: A schematic equal-time correlation function $C(\mathbf{r}, \tau = 0)$ at a temperature just above and just below the critical temperature T_c. At $\mathbf{r} = \mathbf{0}$ the correlation function is $\langle S^2 \rangle = 1$. The distance $\xi(T)$ after which the correlation function decays exponentially to its long-distance value (zero or m^2) is the correlation length. At T_c the correlation length diverges, leading to fluctuations on all scales.

At T_c the equal-time correlation function decays as a power law

at long distances, representing the fact that there are correlations at all length scales (a compromise between short- and infinite-range order). Similarly, the (equal-position) spin-spin correlation function $C(0,\tau) = \langle s(t)s(t+\tau)\rangle$ at long times τ goes to zero for $T > T_c$, to m^2 for $T < T_c$, and at T_c decays as a power law with a different exponent

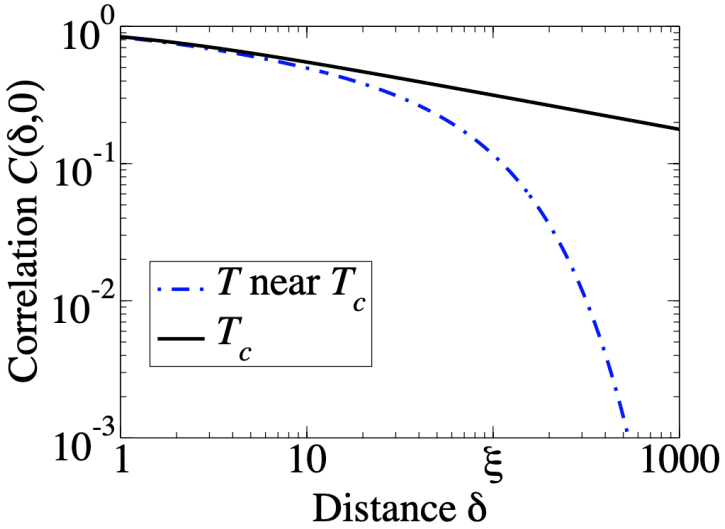
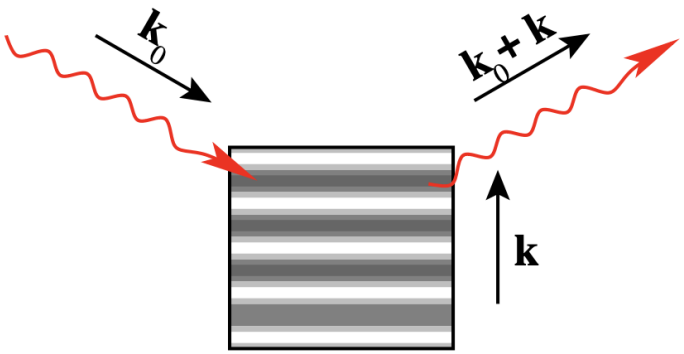


Figure: The schematic correlation function on a log–log plot, both at T_c (straight line, representing the power law $C \sim r^{-(d-2+\eta)}$) and above T_c (where the dependence shifts to $C \sim e^{-r/\xi(T)}$ at distances beyond the correlation length $\xi(T)$).

In other systems, one might study the atomic density–density correlation functions $C(\mathbf{r}, \tau) = \langle \rho(\mathbf{x} + \mathbf{r}, t + \tau)\rho(\mathbf{x}, t)\rangle$, or the height-height correlation function for a surface, or the phase-phase correlations of the superfluid order parameter, etc.

Experimental probes of correlations

Many scattering experiments directly measure correlation functions. In X-ray diffraction a plane-wave beam of wavevector $\mathbf{k}_0$ scatters off of the sample, with the emitted radiation along wavevector $\mathbf{k}_0 + \mathbf{k}$ proportional to $\tilde{\rho}_{\mathrm{e}}(\mathbf{k})$, the Fourier transform of the electron density $\rho_{\mathrm{e}}(\mathbf{x})$ in the sample. The intensity of the scattered beam $|\tilde{\rho}_{\mathrm{e}}(\mathbf{k})|^2$ can be measured, for example, by exposing photographic film. But this intensity is given by the Fourier transform of the equal-time electron density-density correlation function $C_{\mathrm{ee}}(\mathbf{r}) = \langle \rho_{\mathrm{e}}(\mathbf{x}+\mathbf{r}, t)\rho_{\mathrm{e}}(\mathbf{x}, t)\rangle$



Example 2.4: Scaling and corrections to scaling

Sethna problem 12.2

Extra Notes